Introduction

Conducted a preliminary sales data analysis for Adventure Works in preparation for a product launch. The project covered three core analyst tasks: identifying relevant stakeholders, evaluating and sourcing data, and cleaning the dataset for reliable analysis.

Step 1: Identify the stakeholders

1. Briefly explain why any two of the stakeholder groups at Adventure Works may have an interest in the insights related to the sales performance of various products.

Several internal teams at Adventure Works have a stake in understanding sales performance. The sales team needs it to refine their approach and identify what's working ahead of the launch. The marketing team uses it to measure how effective their campaigns are and shape future ones. The product development team looks at it to see how existing products are landing in the market and to make informed decisions around pricing, production, and resource allocation for the new line.

2. Which of the stakeholders need to know about customer preferences when it comes to Adventure Works' products? Why would this information be important to them in relation to the product launch?

When it comes to customer preferences specifically, all four teams benefit. Marketing uses it to sharpen their targeting and messaging. Sales uses it to better match customers with the right products. Product development uses it to ensure new products are built around what customers actually want. Customer service uses it to offer more personalized support.

Step 2: Identify and consider the data sources

1. In your own words, what kind of information can you gather from the provided dataset?

The dataset covers three main areas. Product details: including category, price, weight, and size, help track how individual products perform and hint at what customers gravitate toward. Order details: such as dates, quantities, totals, and statuses, allow for tracking sales over time and spotting any workflow issues. Customer details, particularly the customer ID and payment method fields, help build a picture of repeat behaviour and payment preferences.

2. How can you use the data fields provided to understand sales trends and customer preferences? Briefly discuss your suggested approach for gaining insight into these two aspects separately.

To understand sales trends, grouping order dates, quantities, and totals by product reveals patterns in popularity over time. For customer preferences, looking at which categories, sizes, and payment methods appear most frequently gives a clearer picture of buying behaviour.

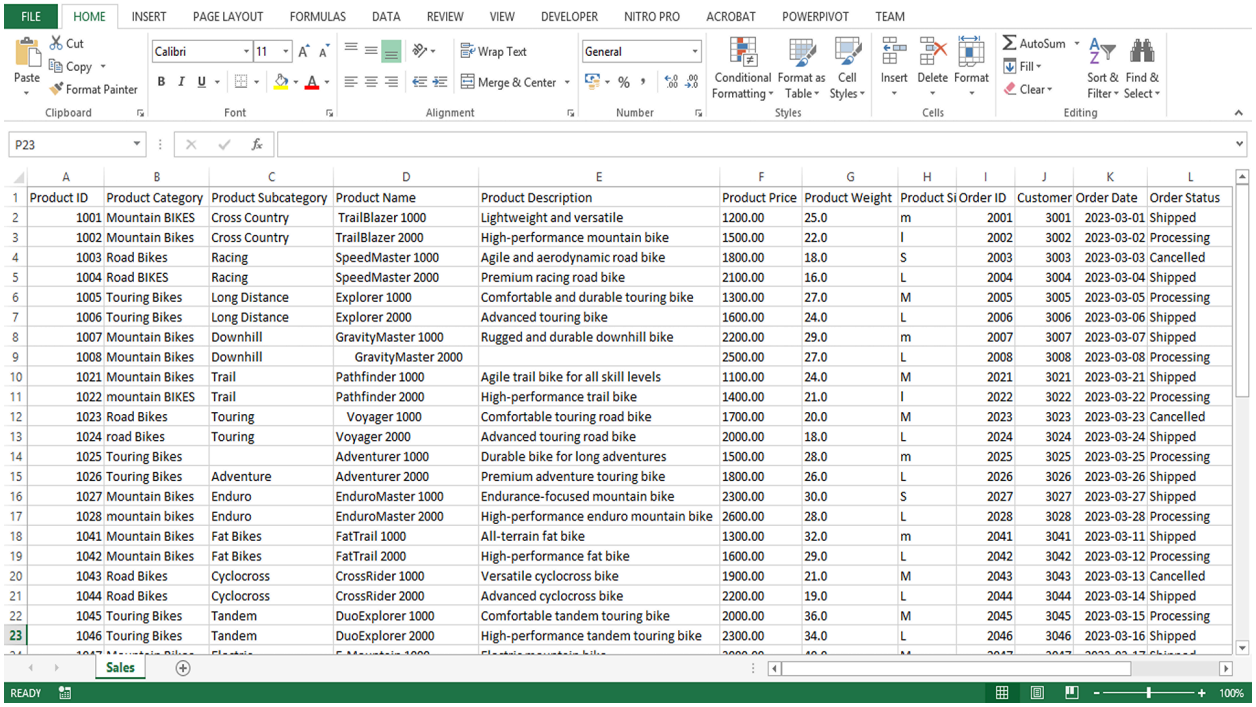
3. Apart from the provided dataset, what other data sources within Adventure Works might be relevant to your analysis or add value to your insights?

Beyond the dataset itself, additional sources could add significant value: customer demographic data, marketing campaign records, supply chain and inventory data, customer reviews, and website analytics would all help build a more complete and actionable analysis.

Step 3: Data import and cleaning

1. List and briefly describe a common Excel function that can help you clean each of the following columns:

Applied TRIM() to remove inconsistent spacing in product names, PROPER() to standardize product category casing, and UPPER()/LOWER() to normalize product size entries.



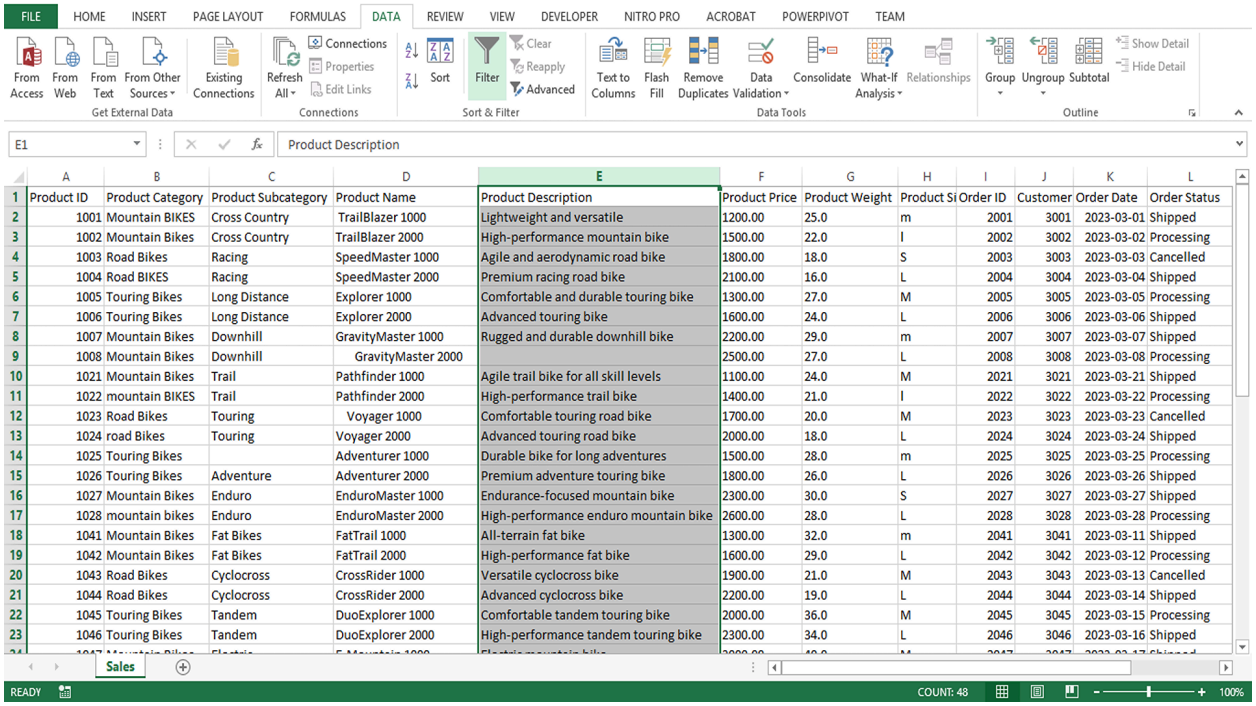
Product ID	Product Category	Product Subcategory	Product Name	Product Description	Product Price	Product Weight	Product Si	Order ID	Customer	Order Date	Order Status
1001	Mountain BIKES	Cross Country	TrailBlazer 1000	Lightweight and versatile	1200.00	25.0	m	2001	3001	2023-03-01	Shipped
1002	Mountain Bikes	Cross Country	TrailBlazer 2000	High-performance mountain bike	1500.00	22.0	l	2002	3002	2023-03-02	Processing
1003	Road Bikes	Racing	SpeedMaster 1000	Agile and aerodynamic road bike	1800.00	18.0	S	2003	3003	2023-03-03	Cancelled
1004	Road BIKES	Racing	SpeedMaster 2000	Premium racing road bike	2100.00	16.0	L	2004	3004	2023-03-04	Shipped
1005	Touring Bikes	Long Distance	Explorer 1000	Comfortable and durable touring bike	1300.00	27.0	M	2005	3005	2023-03-05	Processing
1006	Touring Bikes	Long Distance	Explorer 2000	Advanced touring bike	1600.00	24.0	L	2006	3006	2023-03-06	Shipped
1007	Mountain Bikes	Downhill	GravityMaster 1000	Rugged and durable downhill bike	2200.00	29.0	m	2007	3007	2023-03-07	Shipped
1008	Mountain Bikes	Downhill	GravityMaster 2000		2500.00	27.0	L	2008	3008	2023-03-08	Processing
1021	Mountain Bikes	Trail	Pathfinder 1000	Agile trail bike for all skill levels	1100.00	24.0	M	2021	3021	2023-03-21	Shipped
1022	mountain BIKES	Trail	Pathfinder 2000	High-performance trail bike	1400.00	21.0	l	2022	3022	2023-03-22	Processing
1023	Road Bikes	Touring	Voyager 1000	Comfortable touring road bike	1700.00	20.0	M	2023	3023	2023-03-23	Cancelled
1024	road Bikes	Touring	Voyager 2000	Advanced touring road bike	2000.00	18.0	L	2024	3024	2023-03-24	Shipped
1025	Touring Bikes		Adventurer 1000	Durable bike for long adventures	1500.00	28.0	m	2025	3025	2023-03-25	Processing
1026	Touring Bikes	Adventure	Adventurer 2000	Premium adventure touring bike	1800.00	26.0	L	2026	3026	2023-03-26	Shipped
1027	Mountain Bikes	Enduro	EnduroMaster 1000	Endurance-focused mountain bike	2300.00	30.0	S	2027	3027	2023-03-27	Shipped
1028	mountain bikes	Enduro	EnduroMaster 2000	High-performance enduro mountain bike	2600.00	28.0	L	2028	3028	2023-03-28	Processing
1041	Mountain Bikes	Fat Bikes	FatTrail 1000	All-terrain fat bike	1300.00	32.0	m	2041	3041	2023-03-11	Shipped
1042	Mountain Bikes	Fat Bikes	FatTrail 2000	High-performance fat bike	1600.00	29.0	L	2042	3042	2023-03-12	Processing
1043	Road Bikes	Cyclocross	CrossRider 1000	Versatile cyclocross bike	1900.00	21.0	M	2043	3043	2023-03-13	Cancelled
1044	Road Bikes	Cyclocross	CrossRider 2000	Advanced cyclocross bike	2200.00	19.0	L	2044	3044	2023-03-14	Shipped
1045	Touring Bikes	Tandem	DuoExplorer 1000	Comfortable tandem touring bike	2000.00	36.0	M	2045	3045	2023-03-15	Processing
1046	Touring Bikes	Tandem	DuoExplorer 2000	High-performance tandem touring bike	2300.00	34.0	L	2046	3046	2023-03-16	Shipped

2. With reference to the Adventure Works sales dataset, indicate the most appropriate Excel data type for the content of each column.

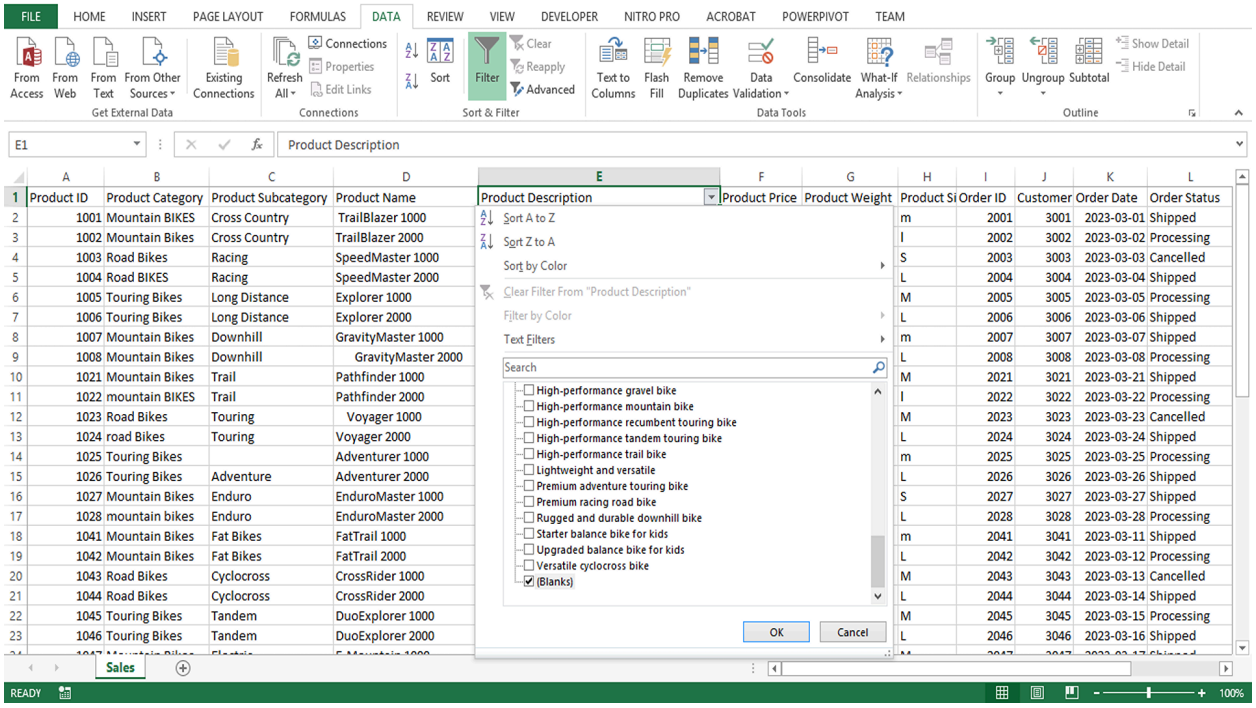
Column	Data Type	Reason

Product ID	Number (0 decimals)	Unique whole number identifier
Product Category	Text	Descriptive label
Product Subcategory	Text	Descriptive label
Product Name	Text	String describing the product
Product Description	Text	Longer descriptive string
Product Price	Currency	Includes decimal values e.g. $19.99
Product Weight	Number (2 decimals)	Includes fractional values e.g. 1.25kg
Product Size	Text	Mixed formats e.g. S, M, L
Order ID	Number (0 decimals)	Unique whole number identifier
Customer ID	Number (0 decimals)	Unique whole number identifier
Order Date	Date	Requires sorting and time-based analysis
Order Status	Text	Descriptive labels e.g. Completed, Pending
Order Quantity	Number (0 decimals)	Whole units, no fractions
Order Total	Currency	Includes decimal values and currency symbols
Payment Method	Text	Descriptive labels e.g. Credit Card

3. Explain how you would exclude the missing entry in the Product Description column in the *Adventure Works Sales* dataset at the source using Excel.



Product ID	Product Category	Product Subcategory	Product Name	Product Description	Product Price	Product Weight	Product Si	Order ID	Customer	Order Date	Order Status
1001	Mountain BIKES	Cross Country	TrailBlazer 1000	Lightweight and versatile	1200.00	25.0	m	2001	3001	2023-03-01	Shipped
1002	Mountain Bikes	Cross Country	TrailBlazer 2000	High-performance mountain bike	1500.00	22.0	l	2002	3002	2023-03-02	Processing
1003	Road Bikes	Racing	SpeedMaster 1000	Agile and aerodynamic road bike	1800.00	18.0	S	2003	3003	2023-03-03	Cancelled
1004	Road BIKES	Racing	SpeedMaster 2000	Premium racing road bike	2100.00	16.0	L	2004	3004	2023-03-04	Shipped
1005	Touring Bikes	Long Distance	Explorer 1000	Comfortable and durable touring bike	1300.00	27.0	M	2005	3005	2023-03-05	Processing
1006	Touring Bikes	Long Distance	Explorer 2000	Advanced touring bike	1600.00	24.0	L	2006	3006	2023-03-06	Shipped
1007	Mountain Bikes	Downhill	GravityMaster 1000	Rugged and durable downhill bike	2200.00	29.0	m	2007	3007	2023-03-07	Shipped
1008	Mountain Bikes	Downhill	GravityMaster 2000		2500.00	27.0	L	2008	3008	2023-03-08	Processing
1021	Mountain Bikes	Trail	Pathfinder 1000	Agile trail bike for all skill levels	1100.00	24.0	M	2021	3021	2023-03-21	Shipped
1022	mountain BIKES	Trail	Pathfinder 2000	High-performance trail bike	1400.00	21.0	l	2022	3022	2023-03-22	Processing
1023	Road Bikes	Touring	Voyager 1000	Comfortable touring road bike	1700.00	20.0	M	2023	3023	2023-03-23	Cancelled
1024	road Bikes	Touring	Voyager 2000	Advanced touring road bike	2000.00	18.0	L	2024	3024	2023-03-24	Shipped
1025	Touring Bikes		Adventurer 1000	Durable bike for long adventures	1500.00	28.0	m	2025	3025	2023-03-25	Processing
1026	Touring Bikes	Adventure	Adventurer 2000	Premium adventure touring bike	1800.00	26.0	L	2026	3026	2023-03-26	Shipped
1027	Mountain Bikes	Enduro	EnduroMaster 1000	Endurance-focused mountain bike	2300.00	30.0	S	2027	3027	2023-03-27	Shipped
1028	mountain bikes	Enduro	EnduroMaster 2000	High-performance enduro mountain bike	2600.00	28.0	L	2028	3028	2023-03-28	Processing
1041	Mountain Bikes	Fat Bikes	FatTrail 1000	All-terrain fat bike	1300.00	32.0	m	2041	3041	2023-03-11	Shipped
1042	Mountain Bikes	Fat Bikes	FatTrail 2000	High-performance fat bike	1600.00	29.0	L	2042	3042	2023-03-12	Processing
1043	Road Bikes	Cyclocross	CrossRider 1000	Versatile cyclocross bike	1900.00	21.0	M	2043	3043	2023-03-13	Cancelled
1044	Road Bikes	Cyclocross	CrossRider 2000	Advanced cyclocross bike	2200.00	19.0	L	2044	3044	2023-03-14	Shipped
1045	Touring Bikes	Tandem	DuoExplorer 1000	Comfortable tandem touring bike	2000.00	36.0	M	2045	3045	2023-03-15	Processing
1046	Touring Bikes	Tandem	DuoExplorer 2000	High-performance tandem touring bike	2300.00	34.0	L	2046	3046	2023-03-16	Shipped



Used Excel's filter functionality to isolate blank entries in the Product Description column and assessed the best approach for handling missing values rather than deleting them outright.

Conclusion

Completed the foundational stages of a real data analysis workflow from stakeholder mapping to data cleaning, reinforcing that reliable analysis depends on understanding the business context and preparing the data carefully before any calculations begin.